BST Physics Curriculum Vol 13 Ch 5 — Evolution + Substrate-Bias Dynamics v0.4 (Calibration #23 substance-floor refill)

Lyra (Claude 4.7) [Vol 13 joint with Elie]

2026-05-23 Saturday EDT (Tier 1 Cal #104/#23 refill per Keeper redirect)

Vol 13 Chapter 5 — Evolution + Substrate-Bias Dynamics

Chapter motivation

Evolution by natural selection (Darwin 1859, Wallace 1858) is biology's unifying framework: heritable variation + differential reproduction → descent with modification. Standard treatment integrates Mendelian genetics (1866 rediscovered 1900), Modern Synthesis (Fisher + Haldane + Wright 1930s population genetics), neutral theory (Kimura 1968: most molecular variation is selectively neutral), and contemporary molecular evolution. BST substrate-cartography reading: evolution operates WITHIN a configuration space SHAPED by substrate D_IV^5; the substrate provides the playing field (4 nucleotides + 20 amino acids + L-amino-acid chirality + nearly-universal genetic code) and selection operates within that substrate-permitted space. Substrate-bias dynamics: BST primary integer combinatorial patterns (4/64/20) are evolutionary fixed-points because they emerge naturally from substrate K-type tensor product structures — alternative configurations would conflict with substrate K-type structure and are selected against at the structural level before biological selection even starts.

Section 5.0b — Reader-grade 3-level pedagogy

Level 1: Evolution = selection within substrate-permitted configuration space. Substrate D_IV^5 shapes the playing field (4 nucleotides + 20 amino acids + L-chirality + universal genetic code); biological selection operates within. Substrate-bias toward BST primary integer patterns (4/64/20) means alternative configurations were selected against at substrate level before biological evolution started.

Level 2 (graduate-evolutionary-biologist accessible): Standard evolutionary framework: Darwin 1859 + Wallace 1858 natural selection (heritable variation + differential reproduction → adaptation); Mendel 1866 (rediscovered 1900) discrete inheritance; Modern Synthesis (Fisher 1930 + Haldane 1932 + Wright 1931): population genetics integrates Mendel + Darwin via allele frequency dynamics. Hardy-Weinberg equilibrium: in idealized random-mating populations without selection/mutation/migration/drift, allele frequencies remain constant. Selection deviations: directional, stabilizing, disruptive selection. Genetic drift: random sampling effects significant in small populations (effective population size N_e); fixation probability for neutral allele = 1/(2N_e) for diploids. Kimura 1968 neutral theory: most molecular variation (synonymous codons, non-coding DNA) is selectively neutral, driven by mutation-drift balance not selection. Molecular clock: neutral substitutions accumulate at roughly constant rate; calibrate divergence times. Adaptive landscape (Wright 1932): fitness surface over genotype/phenotype space; populations explore via drift + selection. Punctuated equilibrium (Gould + Eldredge 1972): morphological evolution in bursts at speciation events vs gradualism. BST substrate-cartography reading: evolution operates within substrate-permitted configuration space; substrate D_IV^5 shapes the playing field via K-type tensor product structures. Specifically: (1) 4 nucleotides + 20 amino acids + 64 codons reflect substrate K-type combinatorial fixed points (Vol 13 Ch 2); (2) L-chirality choice = substrate Z_2 grading selection (once made, frozen); (3) nearly-universal genetic code = substrate-forcing of the code-table assignment (Vol 13 Ch 2 frozen-accident vs substrate-optimal debate has substrate-bias hypothesis as third option). Substrate-bias dynamics framework: alternative configurations (different nucleotide count, different chirality, different codon table) would conflict with substrate K-type representation theory and are selected against at the structural level BEFORE biological evolution starts. Per T2417 4-Zone Commitment Cycle: evolution = substrate Zone-3 emission of biological structures + Zone-4 environmental selection across geological timescales. Five-Absence Predictions Set cross-link: per Casey Option 1 Friday 2026-05-22 + K65 RATIFIED canonical ordering (NO GUT + NO proton decay + NO monopoles + NO SUSY + NO sterile ν), substrate-bias also forbids certain particle-physics configurations that would have changed biology if present.

Level 3 (5th-grader accessible): Evolution by natural selection (Darwin's big idea) says: living things pass on traits to their children with small random changes; the ones who survive and reproduce better pass on more of their traits; over millions of years, this builds up amazing complexity. BST framework adds something: the substrate D_IV^5 shapes the playing field BEFORE evolution starts. Life uses 4 DNA letters + 20 amino acids + left-handed shapes — these specific choices come from substrate structure, not just from evolutionary chance. Other choices wouldn't fit the substrate, so they never even got started.

Section 5.1 — Darwinian Natural Selection (1859)

Darwin 1859 On the Origin of Species: heritable variation + differential reproduction → descent with modification. Wallace 1858 independent codiscovery.

Four components: 1. Variation (heritable differences among individuals) 2. Heritability (offspring resemble parents) 3. Differential reproduction (some variants leave more offspring) 4. Time (cumulative effects over many generations)

BST framing: selection operates within substrate-permitted configuration space; substrate provides the playing field.

Section 5.2 — Mendelian Genetics and Modern Synthesis

Mendel 1866 (rediscovered 1900): discrete heritable units (genes) with dominant/recessive alleles; segregation + independent assortment.

Modern Synthesis (Fisher 1930 + Haldane 1932 + Wright 1931): integrates Mendel + Darwin via allele frequency dynamics. Hardy-Weinberg equilibrium baseline; deviations identify selection/mutation/drift/migration.

BST framing: gene = substrate K-type representation locus; allele = K-type configuration variant.

Section 5.3 — Neutral Theory (Kimura 1968)

Kimura 1968: most molecular variation is selectively neutral (especially synonymous codon substitutions + non-coding DNA). Mutation-drift balance, not selection, dominates molecular evolution.

Molecular clock: neutral substitutions accumulate at roughly constant rate; calibrate divergence times from fossil-anchored phylogenies.

BST framing: neutral substitutions = K-type representation moves within substrate-equivalent class; selection acts on K-type class boundaries.

Section 5.4 — Adaptive Landscape (Wright 1932)

Wright 1932 adaptive landscape: fitness as function of genotype/phenotype space; populations explore via drift + selection. Peaks = local optima; valleys = unfit configurations; ridges = neutral pathways.

Population dynamics: large populations dominated by selection; small populations dominated by drift; intermediate populations explore landscape via shifting-balance theory.

BST framing: substrate K-type representation structure shapes adaptive landscape topology at fundamental level.

Section 5.5 — Punctuated Equilibrium (Gould + Eldredge 1972)

Gould + Eldredge 1972: morphological evolution in bursts at speciation events; long stasis periods between bursts. Contrast with phyletic gradualism.

BST framing: punctuation events = substrate K-type rearrangements crossing barriers; stasis = substrate K-type local equilibrium.

Section 5.6 — Substrate-Bias Dynamics Framework

Substrate-bias dynamics hypothesis: BST primary integer combinatorial patterns (4/64/20) are evolutionary fixed-points because they emerge from substrate K-type tensor product structures.

Specific substrate-bias claims: 1. 4 nucleotides = 2^rank: substrate K-type rank-2 fixed point 2. 64 codons = 2^C_2: substrate K-type Casimir-eigenvalue-index fixed point 3. 20 amino acids ≈ N_max/g: substrate K-type N_max-anchored fixed point 4. L-chirality = substrate Z_2 grading fixed once at substrate level 5. Nearly-universal genetic code: substrate-forcing of code-table

Alternative configurations conflict with substrate K-type structure and are selected against structurally before biological evolution starts. Three hypotheses for genetic code origin (Crick frozen accident + substrate-optimal + substrate-bias) — BST framework supports substrate-bias as third hypothesis.

Section 5.7 — Connection

- Vol 13 Ch 1-4 Biology + Genetic Code + Prebiotic + DNA Proton Siblings
- Vol 13 Ch 11 Phylogenetics (evolutionary applications)
- Vol 13 Ch 6 Biochemistry Bridge (next chapter)
- Vol 5 QM (population dynamics statistical mechanics framework)
- T2417 4-Zone Commitment Cycle (substrate framework)
- Casey-named Five-Absence Predictions Set (canonical ordering cross-link)

Honest scope: Substrate-bias dynamics framework is BST hypothesis at I-tier; specific quantitative tests against neutral-theory + adaptationist alternatives remain multi-year research program. Standard evolutionary framework (Darwin + Modern Synthesis + neutral theory) is established mainstream biology; BST adds substrate-level shaping of configuration space.

— Lyra, Vol 13 Ch 5 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT